Lab 03: Real-World Data Linear Regression

Numerical Methods, Section 3D

Part A: Data

Source: NOAA Global Monitoring Laboratory, Trends in Atmospheric Carbon Dioxide (Mauna Loa Observatory, Hawaii). Full data URL: https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.csv (also viewable at https://gml.noaa.gov/ccgg/trends/)

Description: The Mauna Loa Observatory in Hawaii has continuously measured the concentration of carbon dioxide (CO2) in the atmosphere since 1958. NOAA publishes the annual mean CO2 concentration for each year, computed as the arithmetic mean of that year's monthly averages. This dataset is the longest continuous instrumental record of atmospheric CO2 and is widely used as an indicator of global climate change.

Variables: Independent variable x = Year (calendar year). Dependent variable y = Annual mean atmospheric CO2 concentration, in parts per million (ppm). The 20 most recent complete years (2006-2025) were used.

Year (x)	CO2 (ppm) (y)
2006	382.09
2007	384.02
2008	385.83
2009	387.64
2010	390.10
2011	391.85
2012	394.06
2013	396.74
2014	398.81
2015	401.01
2016	404.41
2017	406.76
2018	408.72
2019	411.65
2020	414.21
2021	416.41
2022	418.53
2023	421.08
2024	424.61
2025	427.35

Part B: Python Script

```
"""
Lab 03: Real-World Data Linear Regression
Data: NOAA Mauna Loa Observatory annual mean atmospheric CO2 concentration
Source: NOAA Global Monitoring Laboratory
        https://gml.noaa.gov/ccgg/trends/  (data file: co2_annmean_mlo.csv)
x = Year
y = Annual mean atmospheric CO2 concentration (ppm)

This script computes a simple linear least-squares regression
(y = a0 + a1*x) "by hand" using the standard formulas, without calling
a built-in regression function, then reports the standard statistics
required for the lab and produces the two required plots.
"""

import numpy as np
import matplotlib.pyplot as plt

# -----------------------------------------------------------------
# 1. Data
# -----------------------------------------------------------------
year = np.array([2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015,
                 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025])

co2 = np.array([382.09, 384.02, 385.83, 387.64, 390.10, 391.85, 394.06, 396.74,
                398.81, 401.01, 404.41, 406.76, 408.72, 411.65, 414.21, 416.41,
                418.53, 421.08, 424.61, 427.35])

n = len(year)

# -----------------------------------------------------------------
# 2. Least-squares coefficients (standard formulas)
#    a1 = ( n*sum(xy) - sum(x)*sum(y) ) / ( n*sum(x^2) - (sum(x))^2 )
#    a0 = ybar - a1*xbar
# -----------------------------------------------------------------
sum_x = np.sum(year)
sum_y = np.sum(co2)
sum_xy = np.sum(year * co2)
sum_x2 = np.sum(year ** 2)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = (sum_y - a1 * sum_x) / n

print(f"Slope (a1)     = {a1:.5f}  ppm/year")
print(f"Intercept (a0) = {a0:.5f}  ppm")

# -----------------------------------------------------------------
# 3. Goodness of fit: Sr (SSE), r^2, standard error of the estimate
# -----------------------------------------------------------------
y_pred = a0 + a1 * year
residuals = co2 - y_pred

St = np.sum((co2 - np.mean(co2)) ** 2)   # total sum of squares
Sr = np.sum(residuals ** 2)              # residual sum of squares (SSE)

r2 = 1 - Sr / St
syx = np.sqrt(Sr / (n - 2))              # standard error of the estimate

print(f"Sr (SSE)       = {Sr:.5f}")
print(f"r^2            = {r2:.6f}")
print(f"s_y/x          = {syx:.5f}  ppm")

# -----------------------------------------------------------------
# 4. Prediction for a year not in the dataset
# -----------------------------------------------------------------
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```
x_new = 2030
y_new = a0 + a1 * x_new
print(f"Predicted CO2 in {x_new}: {y_new:.2f} ppm")

# ---------------------------------------------------------------
# 5. Plots
# ---------------------------------------------------------------
plt.figure(figsize=(7, 5))
plt.scatter(year, co2, color="tab:blue", label="Observed annual mean CO2")
plt.plot(year, y_pred, color="tab:red",
         label=f"Fit: y = {a0:.2f} + {a1:.3f}x")
plt.xlabel("Year")
plt.ylabel("Atmospheric CO2 concentration (ppm)")
plt.title("Mauna Loa Annual Mean CO2 vs. Year")
plt.legend()
plt.tight_layout()
plt.savefig("fit_plot.png", dpi=150)
plt.close()

plt.figure(figsize=(7, 4))
plt.axhline(0, color="black", linewidth=1)
plt.scatter(year, residuals, color="tab:green")
plt.xlabel("Year")
plt.ylabel("Residual (ppm)")
plt.title("Residual Plot")
plt.tight_layout()
plt.savefig("residual_plot.png", dpi=150)
plt.close()
```

Part C: Regression Results

Regression equation:

y = -4417.57 + 2.3919 x
(where y is CO_2 in ppm and x is the calendar year)

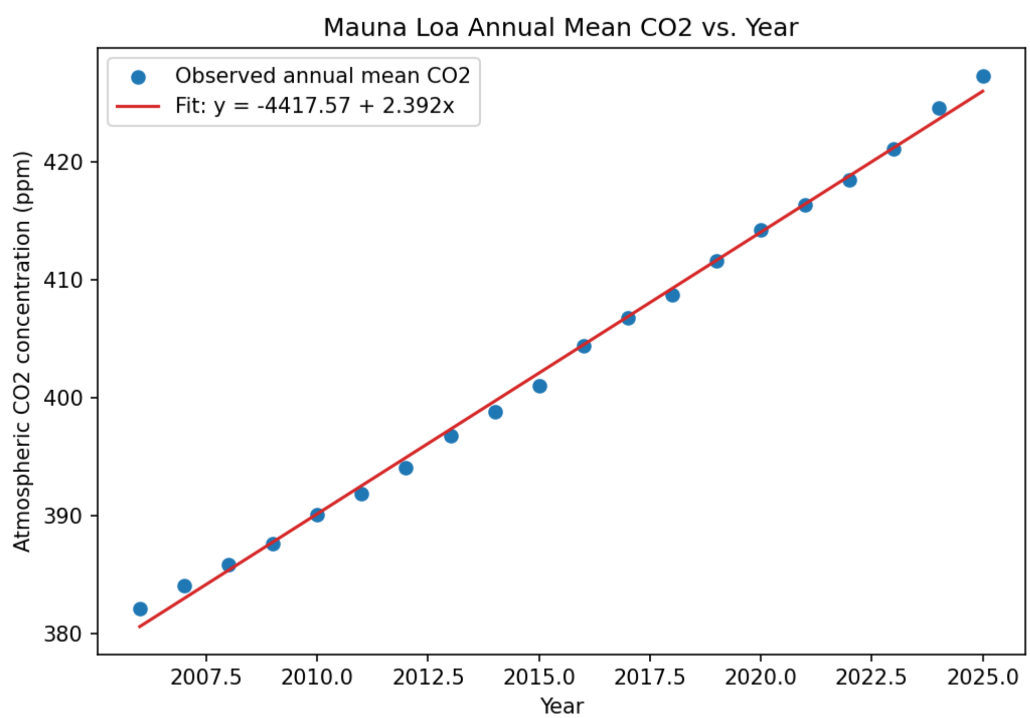
Slope and intercept:

a1 (slope) = 2.3919 ppm/year
a0 (intercept) = -4417.57 ppm (the mathematical y-intercept at year 0; not physically meaningful here since the data only spans 2006-2025)

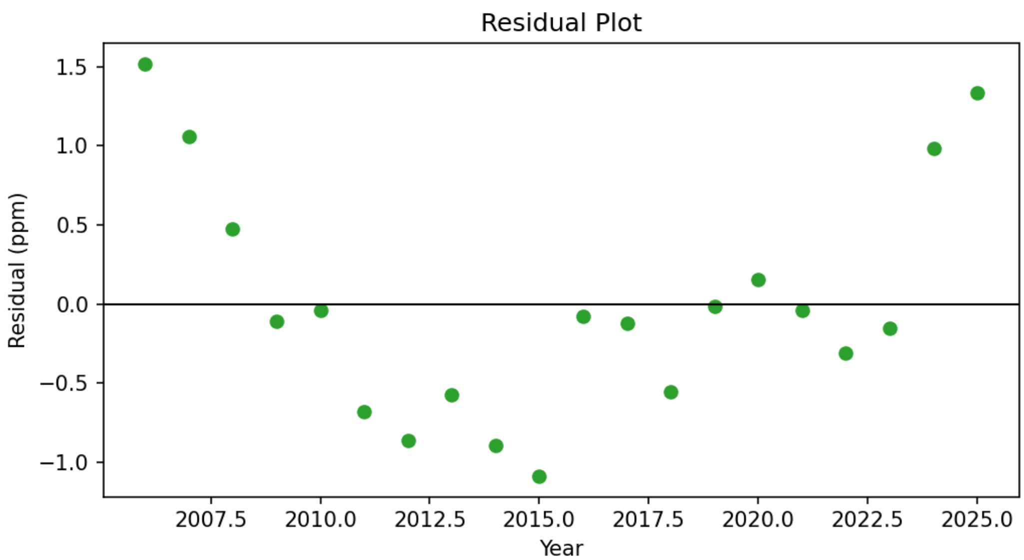
Goodness of fit:

Statistic	Value
Sr (SSE)	10.4072
r-squared	0.99727
Standard error, s(y/x)	0.7604 ppm

Graph of data with fitted line:



Residual plot:


Residual Plot
Residual (ppm)
Year
1.5
1.0
0.5
0.0
−0.5
−1.0
2007.5
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
2025.0

Prediction:

y = -4417.57 + 2.3919(2030) = 437.98 ppm
The model predicts an annual mean atmospheric CO_2 concentration of about 438.0 ppm in 2030, a year outside the fitted data range (2006-2025). This is meaningful as a short-range extrapolation because the underlying trend (fossil-fuel driven CO_2 increase) has been highly linear over the past two decades, so a value only a few years beyond the data is a reasonable estimate, though it assumes emission trends do not change abruptly.

Interpretation:

Slope: The slope of 2.392 ppm/year means that, on average over 2006-2025, atmospheric CO_2 has increased by about 2.39 ppm every year. This matches the widely reported acceleration of the Keeling Curve in recent decades compared to the roughly 1 ppm/year rate observed in the 1960s.

Intercept: The intercept of -4417.6 ppm has no direct physical meaning by itself; it is simply the value the line would take at year 0, far outside the range of the data, and should not be interpreted as a real CO_2 concentration.

Fit quality: With r-squared = 0.9973, about 99.7% of the year-to-year variation in CO_2 concentration is explained by the linear trend with year. The standard error of 0.76 ppm is small relative to the roughly 45 ppm total rise over the 20-year span, confirming the fit is very tight.

Residuals: The residual plot shows small, patternless scatter of roughly +/-1.5 ppm around zero, with no obvious curvature or trend. This supports the choice of a straight-line model for this 20-year window, though a very slight upward curvature is visible if one looks closely (the growth rate itself has been gradually increasing), suggesting a quadratic fit might do even better over a longer time span.